Murphy-Health

An Estimathon-Style Iterative Benchmark for Longevity LLMs

Caltech Longevity Hackathon • Track 01 • LongevityLLM Benchmarking • May 23, 2026

The Problem

Current longevity LLM benchmarks (incl. *LongeBench*) have three blind spots that let a model look smart without being *calibrated*:

- **Limited topic coverage.** RNA-level aging biology — enzymes, disease/hallmark associations, pathway regulation — is sparsely represented relative to its weight in real aging research.
- **Detection only, no estimation.** Tasks are dominated by binary / multi-class classification. Few measure a model's ability to put a *calibrated number* on an aging biomarker.
- **One-shot, no self-correction.** Existing benchmarks score a single final answer. None test whether a model can *iterate* under partial feedback or *recognize its own mistakes.*

The Solution & Pipeline

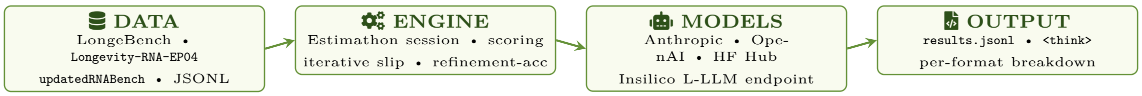

Fig. 1. End-to-end pipeline: open data → Estimathon engine → any model backend → reproducible JSONL.

1. Q-Bank: EP-01 → EP-04 (RNA + Aging) | *open HF*

Contribution. Reproducible RNA-aging suite: **1,295 Qs, 4 formats, 6 RNA types, 13 modifications, 25+ enzymes, 38 pathways, 400+ PubMed refs** — plugs LongeBench's RNA / early-detection gap.

Build. Sarah & Carol seed from primary lit.; Claude does PubMed retrieval, candidate-triple extraction, schema normalization, adversarial filter vs. SotA LLMs; every item human-verified.

Anti-leakage. Monthly job ingests new RNA-aging papers and refreshes a held-out slice — last year's web cannot game next month's eval.

Schema. JSONL fields `{id, format, system, user, choices, answer, pubmed_refs}`; CC-BY-4.0, version-tagged on HF.

ID	Question	Fmt	N
EP-01	ncRNA ↔ aging-disease link	Bin	720
EP-02	RNA-mod → disease class	Tern	231
EP-03	Disease → hallmark of aging	9-MCQ	280
EP-04	Perturb enzyme: up/down/none	Tern	64

2. Algo — Iterative Estimathon

$$S = \left(10 + \sum_{i\in\mathcal{G}} \left\lfloor \frac{u_i}{l_i} \right\rfloor\right) \cdot 2^{\,N-|\mathcal{G}|}$$

Fig. 3a. Scoring fn (Jane Street, 2008); lower S wins.

Mechanic. Each problem gets an interval $[l_i, u_i]$; a slip is **GOOD** iff truth $\in [l_i, u_i]$, **BAD** otherwise — the verdict is the *only* feedback, no direction hint. Shared budget $\lfloor \frac{18}{13} N \rfloor$ slips, cap 3 / problem. Width $\lfloor u/l \rfloor$ punishes hedged $[1, 100]$; $2^{N-|\mathcal{G}|}$ punishes empty problems.

Credibility (not home-grown). Adapted from *Jane Street's Estimathon*, run annually since 2008. The reward / penalty geometry is exactly what quant traders use to score calibrated estimation under partial info:

True Narrow > True Wide ≫ False Wide > False Narrow

This ordering — not the closed form — is what makes the score robust: bet tight *only* when reasons exist; widen *only* when truly uncertain.

Measures. *Refinement accuracy*: random wins 50%, so beating 50% is reasoning, not bracket-search. Captures calibration *and* self-correction in one number.

3. CLI — `pip install murthy-bench`

Quick-start. `pip install murthy-bench` → `murphy run --model opus-4.7 --tasks ep04` — works for Anthropic, OpenAI, Insilico L-LLM, or any HF inference URL.

Stack & features. **Typer** + **Rich** + **prompt_toolkit** drive the UX (typed CLI, live color slips, slash-cmd autocomplete `/test /batch /model /explore`); **anthropic/openai** + **huggingface_hub** + **datasets**; **pydantic**-validated resumable JSONL with `<think>`-trace capture; 8-thread batch; PyPI publish via GitHub Actions + OIDC.

```
$ murphy run --model sonnet-4-6 --tasks ep04

murphy v0.2.1 | anthropic | tasks=20
#01 LB 0010 GOOD w=1 [1.0e7→5.7e6]
#02 LB 0010 BAD  w=1 [5.7e6→5.7e6]
→ INTERVAL [20, 55] (2 left)

slips          3/40 solved          1/20
```

Fig. 3b. Live murphy CLI — Rich-rendered.

EP-04 prompt — exactly what the LLM sees, with rationale

System. You are an RNA-aging expert. Reply with exactly one token from {`promote`, `suppress`, `no_effect`}. No explanation.

User. enzyme = **METTL3** tissue = muscle perturbation = knockdown

Q: Does this perturbation **promote**, **suppress**, or have **no effect on** aging?

Assistant. ______ *(truth: suppress)*

Why the prompt matters. Every Murphy item ships its exact LLM-facing template (system + user + assistant) in the JSONL — no "ambiguous wording" excuse. On a BAD slip the model gets only `BAD`, never "too high" / "too low"; refinement must come from reasoning, not gradient-following. `no_effect` is a real option — catches confidently-wrong models.

Results — Leaderboard

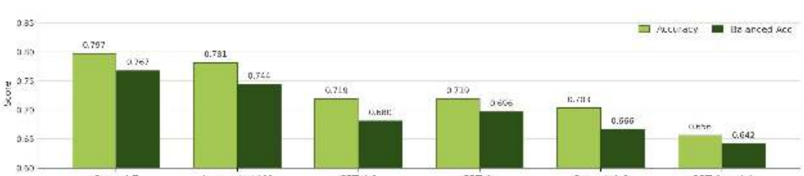

Fig. 4. Leaderboard — 6 LLMs on RNA-aging Estimathon ($n = 20$).

Headline. **Opus-4.7** tops Murphy at **0.797 / 0.767** (acc / bal-acc); Insilico's **LongevityLLM** is a close 2nd (**0.781 / 0.744**) — a domain-tuned model meaningfully outperforms GPT-4.1 / 4o / Sonnet-4.6 / 4o-mini. The acc / bal-acc gap widens as model strength drops — weaker models exploit class imbalance instead of reasoning.

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github.com/OhhMoo/Murphy-Health • carolw/Longevity-RNA-EP04 • sarahliu/updatedRNABench • `pip install murthy-bench`